

Catalytic Asymmetric Synthesis of Zinc Metallacycles

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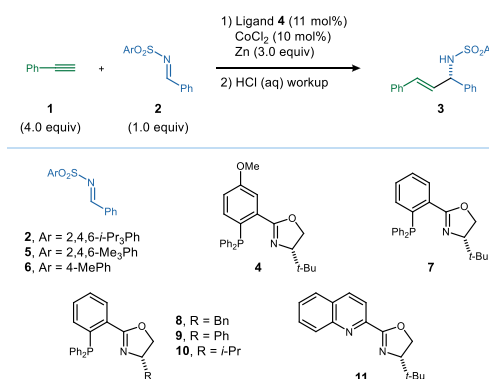
Supporting Information

ABSTRACT: Transition-metal-catalyzed reductive coupling reactions of alkynes and imines are attractive methods for the synthesis of chiral allylic amines. Mechanistically, these reactions involve oxidative cyclization of the alkyne and the imine to generate a metallacyclic intermediate, which then reacts with H₂ or a H₂ surrogate to form the product. As an alternative to this hydrogenolysis pathway, here we show that transmetalation to zinc can occur, forming a zinc metallacycle product. This organozinc product serves as a versatile nucleophile for carbon–carbon and carbon–heteroatom coupling reactions. Mechanistic studies based on isotopic labeling experiments and DFT calculations suggest that the key transmetalation step occurs between a Co(II) species and ZnCl₂.

Transmetalations from organometallic reagents to transition metal catalysts underlie numerous catalytic processes (Figure 1). For example, the Negishi cross-coupling reaction involves a transmetalation step from an organozinc nucleophile to a palladium catalyst.¹ In principle, the reverse process, transmetalation from a catalyst to a main group metal, would provide access to complex organometallic compounds that could function as intermediates for further functionalization. Achieving such a transformation would require careful consideration of the thermodynamics of metal–carbon and metal–halide bonds in order to control the directionality of transmetalation.²

Simple cobalt salts are known to catalyze the synthesis of arylzinc reagents from aryl halides and Zn metal.³ In the

Table 1. Effect of Reaction Parameters



entry	variation from standard conditions ^a	yield (%) ^b	ee (%)
1	none	79	91
2	no CoCl ₂	0	—
3	no ligand 4	0	—
4	2.0 equiv of 1	53	91
5	Mn instead of Zn	0	—
6	imine 5 instead of 2	81	85
7	imine 6 instead of 2	79	45
8	ligand 7 instead of 4	59	88
9	ligand 8 instead of 4	60	59
10	ligand 9 instead of 4	56	62
11	ligand 10 instead of 4	45	39
12	ligand 11 instead of 4	27	38

^aStandard reaction conditions: 1 (0.8 mmol), 2 (0.2 mmol), Zn (0.6 mmol), CoCl₂ (10 mol %), ligand 4 (11 mol %); 24 h, rt, DMA (1 mL). ^bIsolated yields determined after purification by column chromatography.

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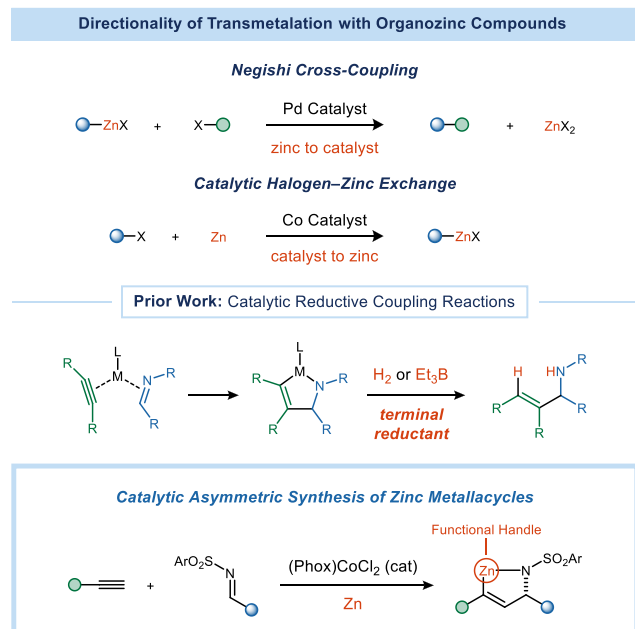
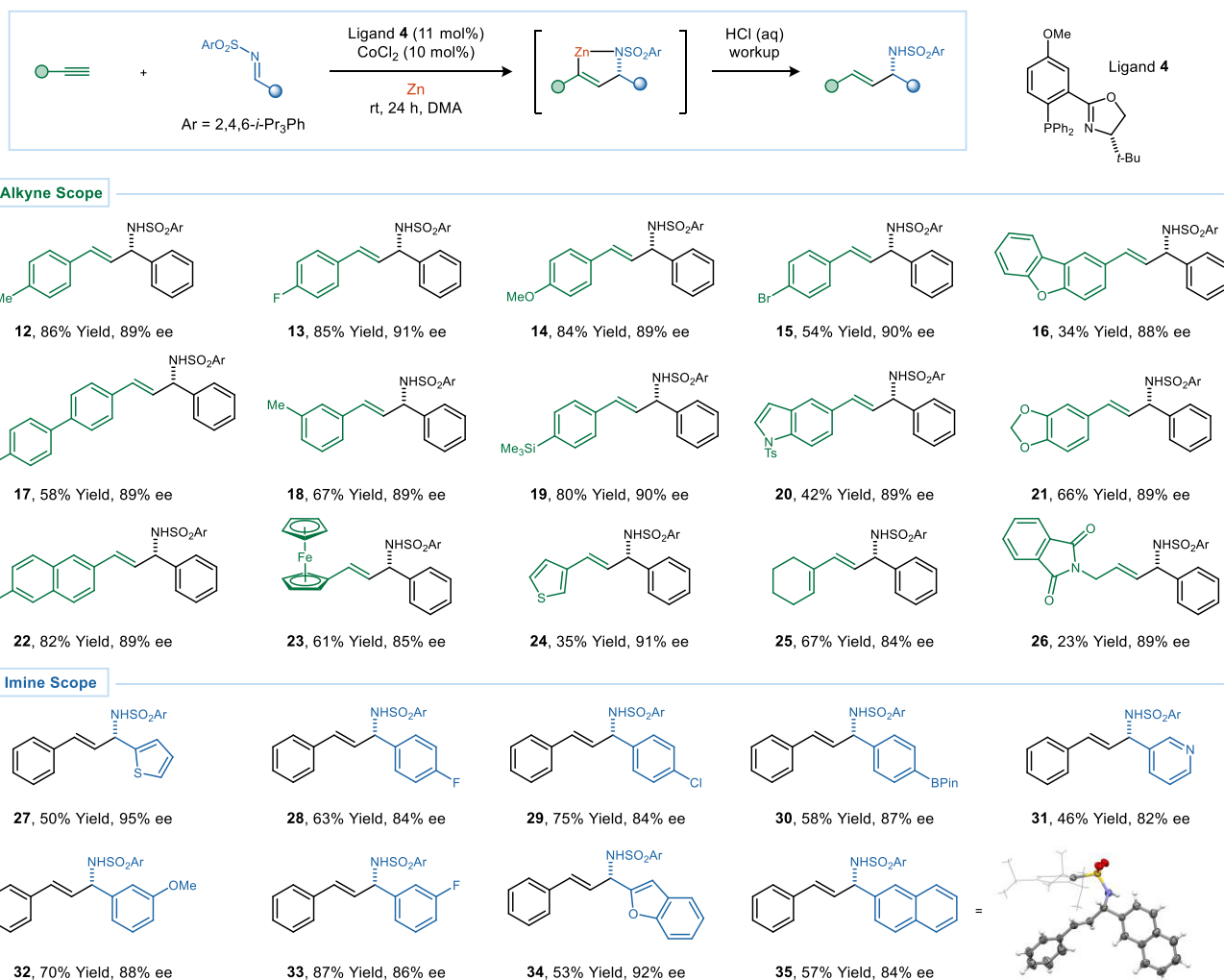


Figure 1. Catalytic asymmetric synthesis of zinc metallacycles.

Table 2. Substrate Scope Studies^{a,b}

^aReaction conditions: **1** (0.8 mmol), **2** (0.2 mmol), Zn (0.6 mmol), CoCl₂ (10 mol %), ligand **4** (11 mol %); 24 h, rt, DMA. Isolated yields determined after purification by column chromatography are reported. Enantiomeric excess (ee) values were determined by HPLC analysis using chiral columns. ^bThe absolute stereochemistry was established by XRD analysis of **35**. All other compounds are assigned by analogy.

proposed mechanism, a reduced Co species activates the aryl halide by oxidative addition. Transmetalation from the Co(II)–aryl intermediate to ZnX₂ generates the organozinc product. Here we show that the favorable thermodynamics of transmetalation from Co(II) to Zn can be successfully merged with reductive C–C coupling⁴ to achieve an asymmetric synthesis of Zn metallacycles. In previous examples of imine and alkyne reductive coupling reactions, a terminal reductant such as H₂ or a H₂ surrogate is used, and the products contain a hydrogen atom at the site of the M–C bond.⁵ The same process has also been achieved in a redox-neutral fashion using an amine and an alkyne through a hydrogen borrowing strategy.⁶ Alternatively, there are limited cases where a third organoboron component can be introduced to carry out tandem oxidative cyclization and C–C cross-coupling.^{7,8} Finally, there is a singular example of transmetalation to form an aluminum metallacycle.⁹ However, the need for Me₃Al limits functional group compatibility, and synthetic elaboration of the Al–C bond is limited to halogenation. By instead transmetalating to Zn, it is possible to incorporate a versatile functional handle to yield a diversity

of allylic amine products¹⁰ containing geometrically defined trisubstituted alkenes.

Under the optimized reaction conditions, a (Phox)CoCl₂ catalyst promoted the reductive coupling of imine **2** and phenylacetylene (**1**) with Zn. During workup, the putative organozinc product was quenched with HCl(aq) to yield allylic amine **3** in 79% yield with 91% ee (Table 1, entry 1). When the crude reaction mixture was analyzed prior to the acidic quench, an intermediate species containing resonances at 6.55 ppm (d, *J* = 3.1 Hz) and 5.16 ppm (d, *J* = 3.1 Hz) was observed in the ¹H NMR spectrum. When PhCCD was used, the peak at 6.55 ppm was absent, and the peak at 5.16 ppm appeared as a singlet. Based on these data, we assign this intermediate as an organozinc species of the general form shown in Figure 1. The small *J* value is consistent with a HCCH dihedral angle of approximately 60°, which would be expected for a metallacyclic rather than an acyclic structure.

No yield of the product is observed in the absence of CoCl₂ or the Phox ligand (entries 2 and 3). Due to a competing alkyne homo-oligomerization process, alkyne **1** is used in excess and added in two portions. Reducing the number of equivalents of the alkyne or adding it in a single

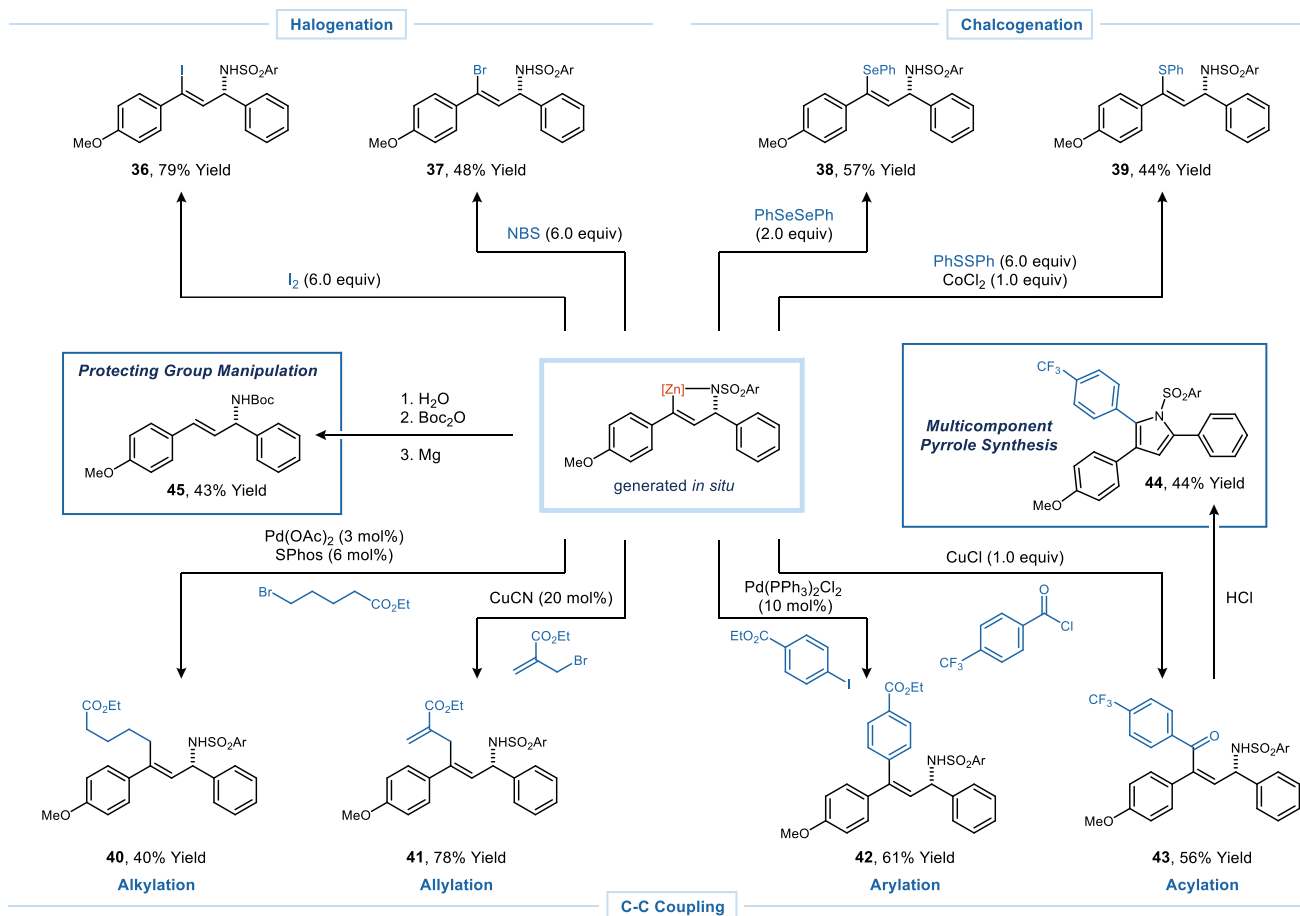


Figure 2. Functionalization experiments of the organozinc intermediate. Yields of isolated products over the two-step sequence of reductive coupling and organozinc functionalization are reported. See the [Supporting Information](#) for experimental details.

portion lowers the yield (entry 4). Mn proved to be ineffective as a reductant (entry 5). A hindered sulfonyl group was critical for enantioselectivity. Less hindered groups provided similar yields but lower ee values (entries 6 and 7). Other Phox or Quinox ligands generated some amount of product but were inferior with respect to yield and ee (entries 8–12).

The substrate scope of the reaction is summarized in [Table 2](#). Aryl, heteroaryl, and ferrocenyl-substituted acetylenes proved to be effective in the reaction. Notably, C(Ar)–Br bonds were tolerated (product 15), despite the ability of CoX₂ to catalyze the synthesis of organozinc compounds from aryl bromides and Zn. An enyne substrate yielded a product containing a conjugated diene (product 25). Finally, alkylacetylenes provided ee values similar to those observed with arylacetylenes but gave diminished yields (product 26). Internal alkynes were ineffective, in striking contrast to other examples of catalytic imine–alkyne reductive coupling reactions, which require internal alkynes and are not viable for terminal alkynes.^{5a–d,7,8} With regard to the imine partner, a variety of aryl and heteroaryl substituents were tolerated. Alkyl imines, on the other hand, proved to be unreactive.

The zinc metallacycles generated in this reaction serve as versatile intermediates for the synthesis of allylic amines containing trisubstituted alkenes ([Figure 2](#)).¹¹ For example, quenching of the reaction with I₂ or *N*-bromosuccinimide (NBS) instead of HCl(aq) yielded halogenated products 36 and 37, respectively. Related chalcogenation reactions were

carried out with PhSeSePh or PhSSPh (products 38 and 39). In the latter case, the addition of CoCl₂ (1.0 equiv) was beneficial for the yield.¹² There is no erosion of the ee during the organozinc functionalization step.

Carbon–carbon cross-coupling reactions were also investigated. Alkyl cross-coupling with a primary alkyl halide electrophile was conducted using an (SPhos)Pd catalyst (product 40). A Cu-catalyzed allylation provided skipped diene product 41.¹³ Negishi cross-coupling yielded product 42 containing a trisubstituted alkene. Due to the metallacyclic nature of the organozinc intermediate, the alkene was obtained as a single geometric isomer despite the minimal differentiation between the two aryl groups. Acylation was achieved in high yield by transmetalation to a copper nucleophile.¹⁴ Treatment of the acylated product 43 with HCl induced cyclization to afford pyrrole product 44. This sequence represents a one-pot, three-component pyrrole synthesis from an alkyne, an imine, and an acyl chloride. Finally, the hindered sulfonyl group on the imine could be replaced with a Boc group by initial treatment with Boc₂O and then deprotection under reducing conditions with Mg.^{6a}

Mechanistic details of the reaction were examined using deuterium-labeling experiments ([Figure 3A](#)). Quenching the reaction with CD₃OD in the glovebox yielded product 3 with 75% deuteration at C3, which is the expected position based on the proposed structure of the zinc metallacycle. Conversely, when PhCCD (92% D) was used as a substrate, 91% deuteration was observed at C2. Interestingly, there was

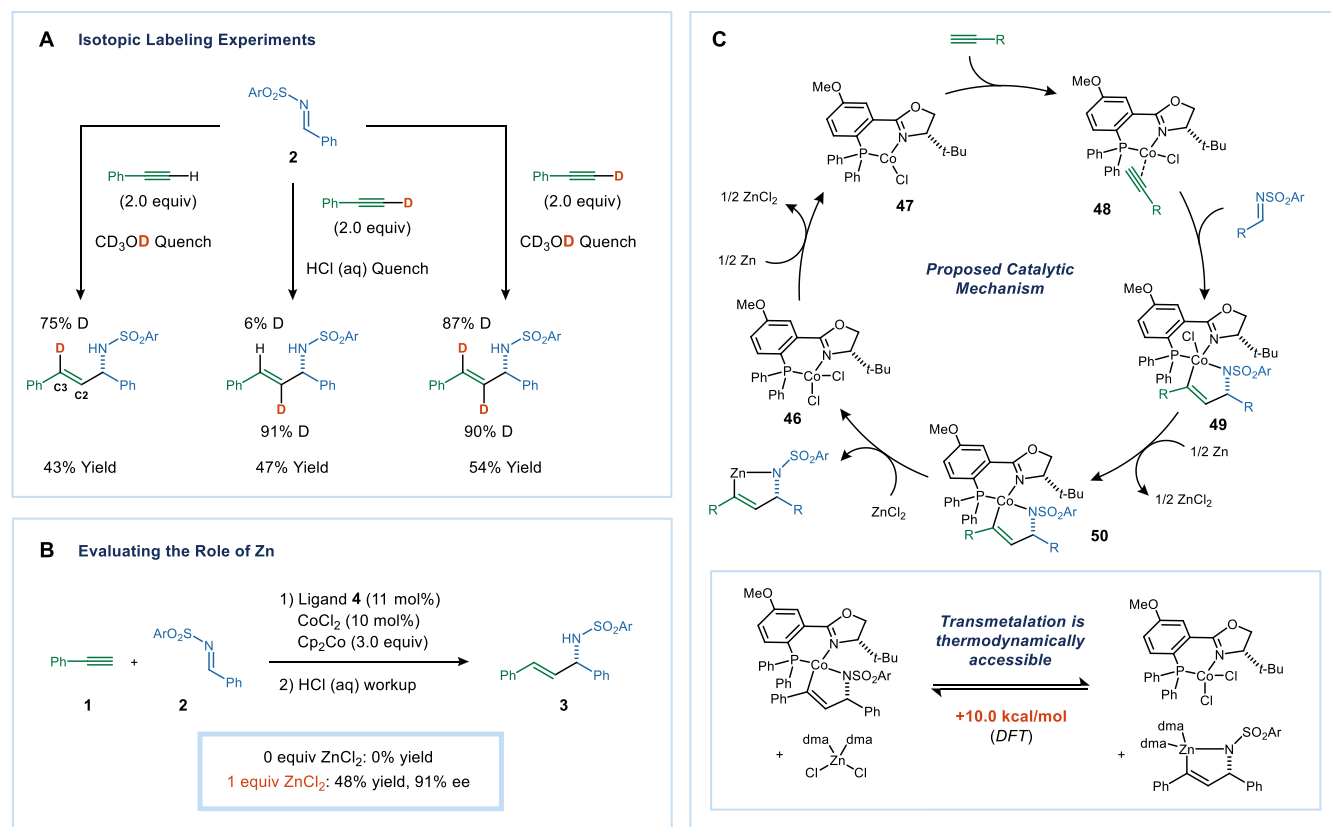


Figure 3. (A) Isotopic labeling experiments probing the structure of the proposed zinc metallacycle intermediate. (B) The dual role of zinc in the catalytic cycle can be replicated with Cp₂Co and ZnCl₂. (C) Proposed catalytic mechanism and DFT calculations for the transmetalation step from Co to Zn (M06-L-D3(SMD)/6-31G(d,p)).

6% deuteration at C3, suggesting that the alkyne is sufficiently acidic to partially protonate the Zn metallacycle during the reaction. Finally, the combination of PhCCD and the CD₃OD quench led to high deuteration at both C2 and C3.

Zn plays a dual role in the catalytic mechanism. It serves as an electron source to turn over the catalyst. It is also required in the transmetalation step to generate the Zn metallacycle product. Accordingly, when Zn was replaced with Cp₂Co, there was no yield of coupled product, despite the fact that Cp₂Co is sufficiently reducing to access the reduced Co(I) state of the catalyst (Figure 3B). Using Cp₂Co in combination with ZnCl₂ restores the reactivity, and product 3 is generated in 48% yield with 91% ee.

A proposed catalytic mechanism is shown in Figure 3C. Reduction of the (Phox)CoCl₂ catalyst 46 generates Co(I) complex 47, which can then induce oxidative coupling of the alkyne and imine to form a Co(III) metallacycle (49).¹⁵ Transmetalation to ZnCl₂ could then occur to form (Phox)CoCl₃ and the product. Alternatively, 49 could be reduced to a Co(II) metallacycle (50) prior to transmetalation. To distinguish between these two possibilities, we carried out DFT calculations to examine the thermodynamics of transmetalation. At the Co(III) oxidation state, transmetalation from Co to Zn is disfavored by 21.2 kcal/mol. By comparison, transmetalation from Co(II) to Zn is thermodynamically more favorable (+10.0 kcal/mol). We note that there is uncertainty regarding the precise nature of the Zn species in solution due to potential aggregation and variable solvent coordination. These issues would impact the

absolute values of the calculated equilibrium constants. However, the relative energies nonetheless suggest that reduction from Co(III) to Co(II) may precede transmetalation.

In summary, reductive coupling reactions of π systems provide an approach to C–C bond formation that takes advantage of readily available starting materials. Typically, the mechanism involves the formation of a metallacyclic intermediate that reacts with a terminal reductant, such as H₂, to form the coupled product. By using Zn as a reductant, it is possible to instead carry out a catalyst-to-Zn transmetalation, which furnishes an organozinc product that is poised for further functionalization. This process avoids the need to generate organozinc compounds using a preinstalled halide or by deprotonation using a strong base. Applications of this transmetalation to other multicomponent reductive coupling reactions are ongoing.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c05885>.

Experimental details, characterization data, and computational details (PDF)

Accession Codes

CCDC 2267033 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cam-

bridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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